

Economics of Cleaner Industrial Boilers in Nepal



Initial Assessments using the iBEAN Tool

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Outline

- **iBEAN Tool**

- Background
- Method
- Input/Output

- **Base case**

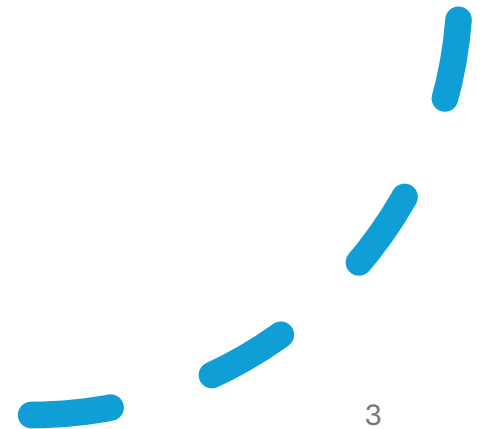
- Selected survey summary
- Boiler purchase cost
- Base case parameters
- Results

- **Sce-rio cases (Incentive Designs)**

- Capital subsidy (Electric)
- Production loan
- Increase in fossil fuel price
- Pellet promotion

iBEAN Tool

- **Background**
- **Methods**
- **Input/Output**



iBEAN Tool

Background

- **industrial Boiler Economic A-lysis for Nepal (iBEAN)**
- **Estimates** the annual and total levelized costs of a new industrial steam boilers
- **Identifies** relevant incentive designs to promote cleaner boilers
- **Open-source** MS Excel-based tool

iBEAN Tool

Methods

Annuity-based Levelized Analysis

$$LCOB = CAPEX + OPEX^* + \left[F_p \times \left(\frac{1}{CV_f} \times \frac{UE}{\eta_b} \right) \right]$$

Where,

LOB is levelized cost (annuity) of boiler;

CAPEX is capital cost calculated using $CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$

OPEX* is annual operating expenditure (labor and others but exclude fuel cost);

Annual fuel cost: $F_p \left(\frac{1}{CV_f} \times \frac{UE}{\eta_b} \right)$

where, **F_p** is fuel price, **CV_f** is gross calorific value of fuel, **UE** is useful energy required per year; **η_b** is rated boiler efficiency.

Total Levelized Analysis

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$$

Where,

NPV is the net present value;

CF_t is the **cash (out)flow** at time *t*,

r is the discount rate;

n is the number of time periods.

- **Externality**
 - Carbon Tax
 - Environmental Damage Cost

Key Boiler Input Parameters

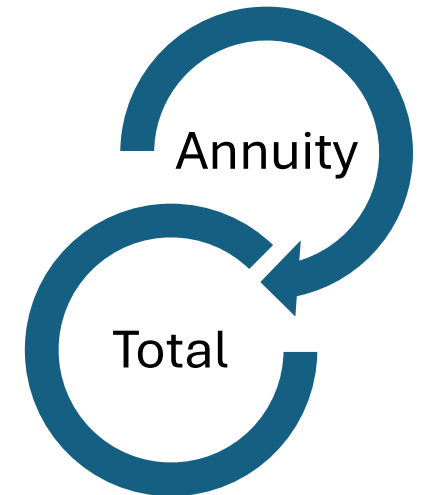
Technical
• Capacity • Lifetime • Operational factors • Efficiency • Calorific value • Fuel/water requirement • Labor requirement • Emission factor

Annuity
• CAPEX • Purchase price • Market interest rate • Loan period • OPEX • <i>Variable costs</i> • Fuel/water price • O&M • <i>Fixed costs</i> • Labor • Installation • Insurance • Contingency

+

Total
• Discount rate • Down payment • Equipment grant • Inflation • PCE (Yes/No) • Carbon tax • Environmental damage cost • PM2.5 • CO2

Output
(levelized)

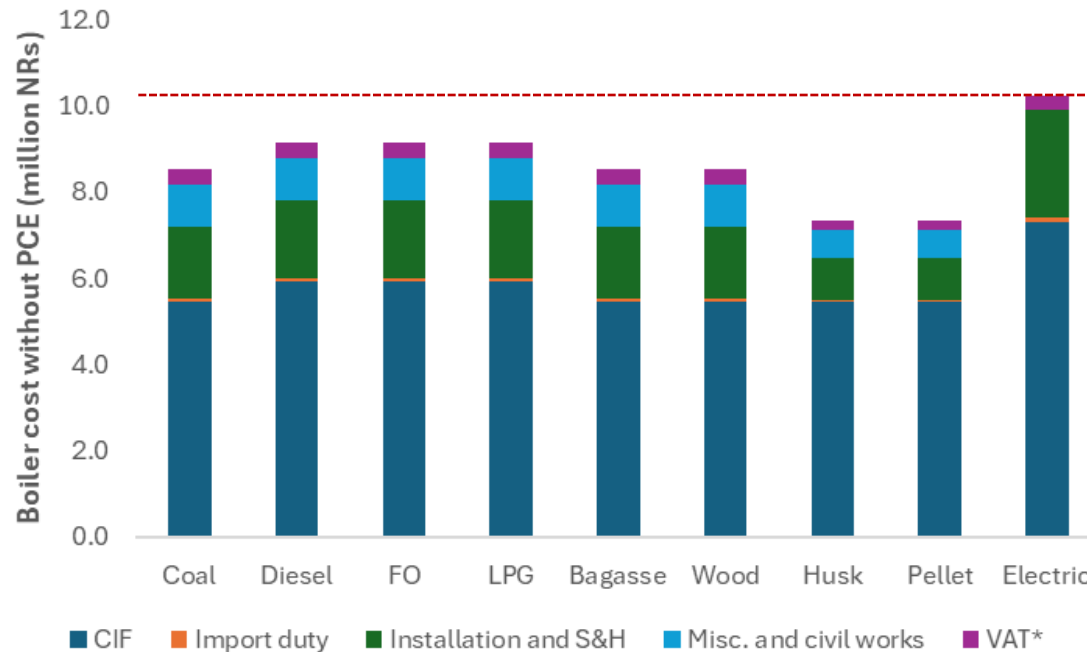


Base Case

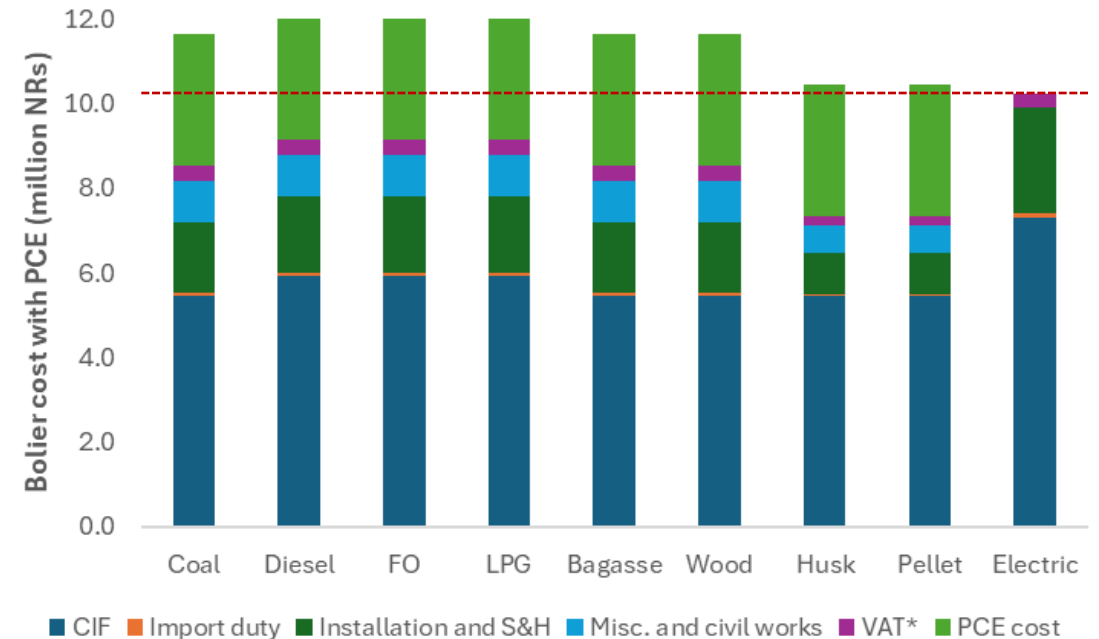
- **Selected survey summary**
- **Boiler purchase cost**
- **Base case parameters**
- **Results**

Boiler (1 tph) Capital Cost

Without Pollution Control Equipment (PCE)



With Pollution Control Equipment (PCE)



- **Electric** boiler is currently the **most expensive** compared to **all other boilers** (fossil-fuel- and biomass-based) **without PCE**. Other boilers cost around **72% to 90%** of the cost of an **electric** boiler.
- **With PCE cost** included, **biomass-based** and **fossil-fuel** based boilers become **more expensive** compared to **electric** boiler. Around **20%** higher for **oil-based** boilers, **14%** higher for **coal**, **bagasse** and **wood**, and **2%** higher for **husk** and **pellet**.

Current Boiler (Electric and Pellet) and Pollution Control Equipment (PCE) Capital Costs

	Electric boiler (1 tph)		Pellet boiler (1 tph)		PCE (bag filter)	
	NRs (million)	USD ('000)	NRs (million)	USD ('000)	NRs (million)	USD ('000)
CIF+IDuty	7.4	52.9	5.5	39.3	2.4	17.5
Others*	2.6	18.5	1.7	12.0	0.6	4.3
VAT**	0.3	2.3	0.2	1.5	0.1	0.6
Total	10.3	73.7	7.4	52.8	3.1	22.3

Notes: * includes shipping and handling, installation, and other miscellaneous and civil works.** VAT is applicable only to others. Exchange rate: 1 USD =140 NRs

Data source: Electric boiler: Thermax quotation. Pellet boiler Cheema (AD Excel). PCE bag filter (Cheema quotation)

- **Other costs** (S&H, Installation, civil works, etc.) for an electric boiler are about **one-third** of the equipment cost.
- **Other costs** for the **pellet** boiler are about **one-fifth** of the equipment cost.
- **Other costs** of **PCE** are about **one-fourth** of the equipment cost.

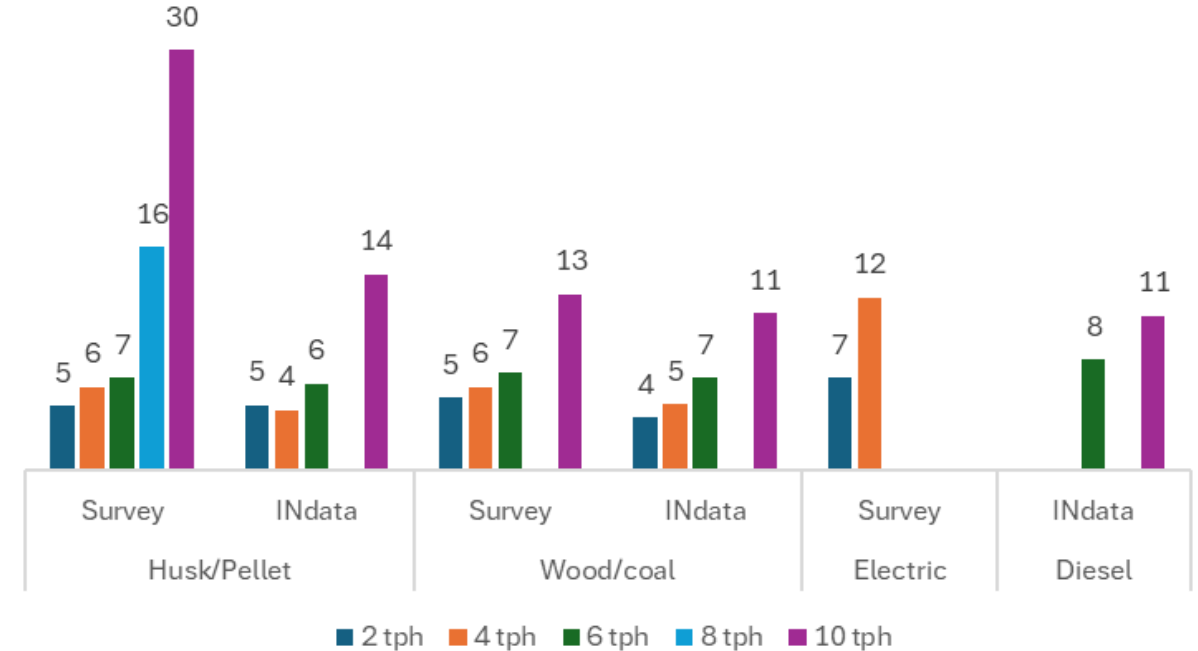
Boiler Purchase Cost with different capacities

Boiler purchase cost* (million NRs)

	Husk/Pellet			Wood/coal			Electric		Diesel	
	Quote	Survey	India	Quote	Survey	India	Quote	India	Quote	India
1 tph	5.50	-	-	5.47	-	-	7.40	-	5.94	-
1.5 tph	6.38	-	-	6.40	-	-	-	-	-	-
2 tph	-	4.67	4.70	-	5.22	3.84	-	6.62	-	-
4 tph	-	6.00	4.35	-	6.00	4.74	-	12.37	-	-
6 tph	-	6.60	6.23	-	7.00	6.60	-	-	-	7.97
10 tph	-	30.00	14.03	-	12.60	11.22	-	-	-	11.06

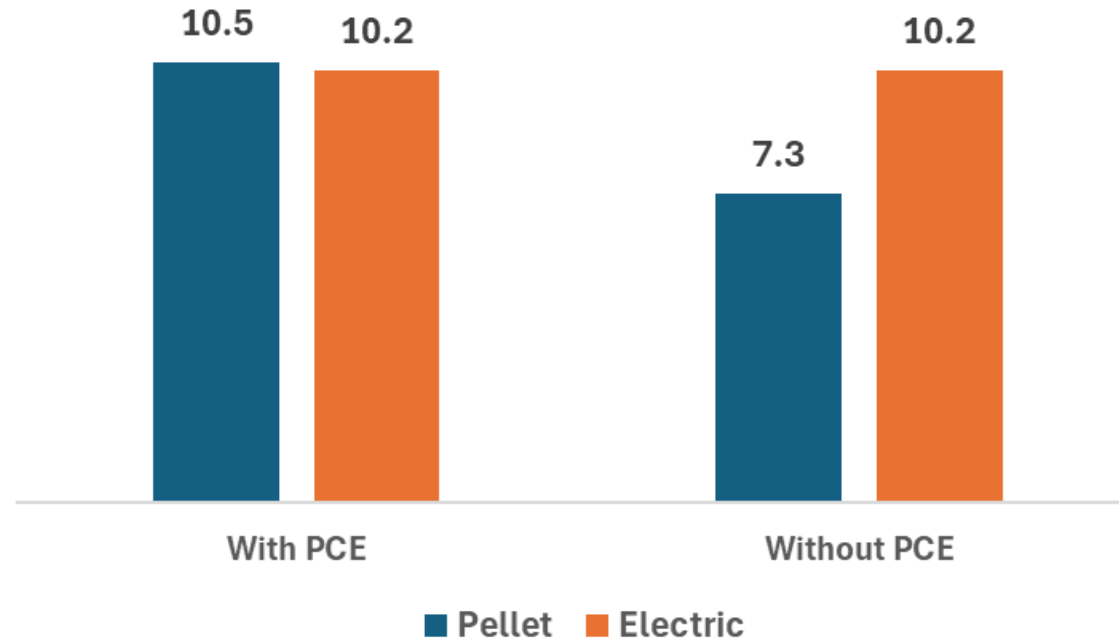
Notes: * Boiler purchase costs exclude installation, transport, civil works, and PCE costs. India data also excludes import duty and VAT. Survey data is only for 2018 (boiler installed year) onwards. - Indicates data not available.

Boiler purchase cost* (million NRs)



- **Electric** boilers are currently **more expensive** than **husk, wood, and diesel** boilers (without pollution control equipment) for different boiler capacities.
- Purchase prices do not vary much for **2 tph** and **4 tph** capacities for **wood** boilers.
- The purchase price of **electric** boilers with capacities of **2 tph** and **4 tph** is almost **double**.

Capital cost (million NRs): Electric vs Pellet Boiler



- The **capital price** of an **electric** boiler is **higher** compared to a **pellet** boiler **without** Pollution Control Equipment (**PCE**).
- **With PCE**, the **capital cost** of a **pellet** boiler is **more than** that of an **electric** boiler.

Base case parameters

Lifetime (year)	20
Discount rate	6%
Loan	Yes
Loan term (year)	10
Down payment	0%
Pollution control	Yes
Inflation rate*	5.5%
Interest rate	8%

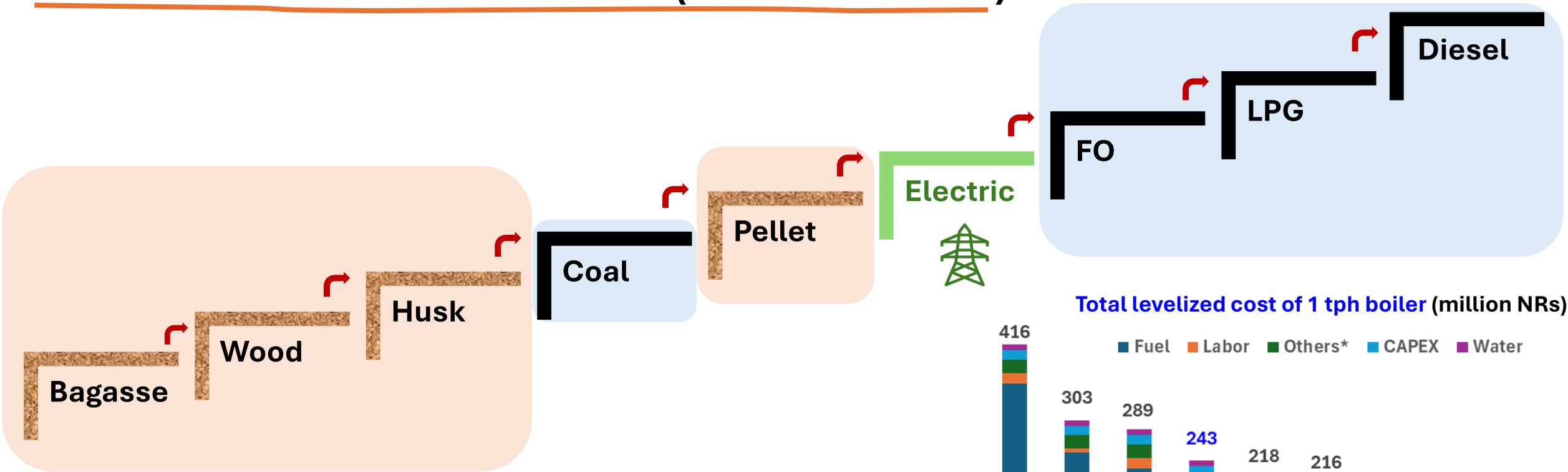
* Excluded in fuel costs and carbon taxes.

Fuel price

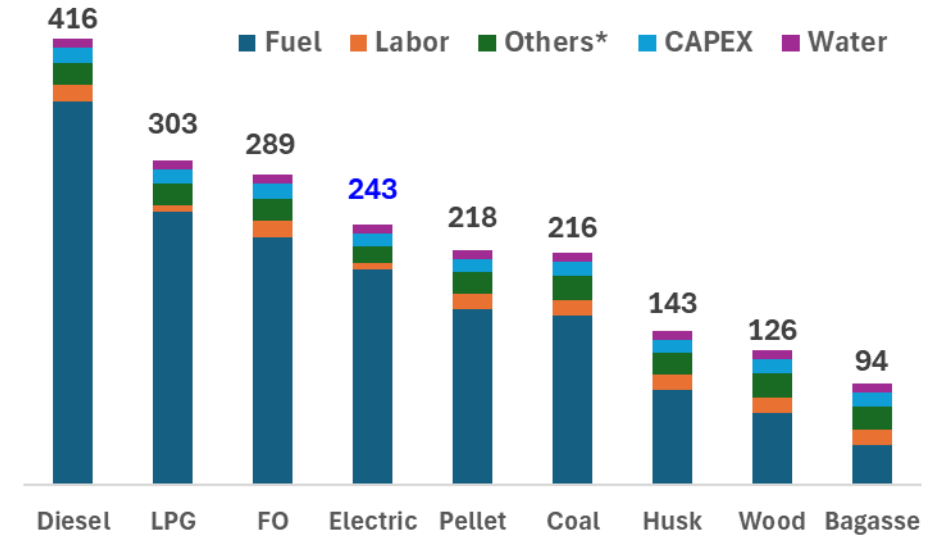
Coal	40
Diesel	181
FO	113
LPG	135
Bagasse	8
Wood	10.7
Husk	12.9
Pellet	32.5
Electric+	10

+ NRs/kWh, others are in NRs/kg.

Levelized Total Cost (Base case)



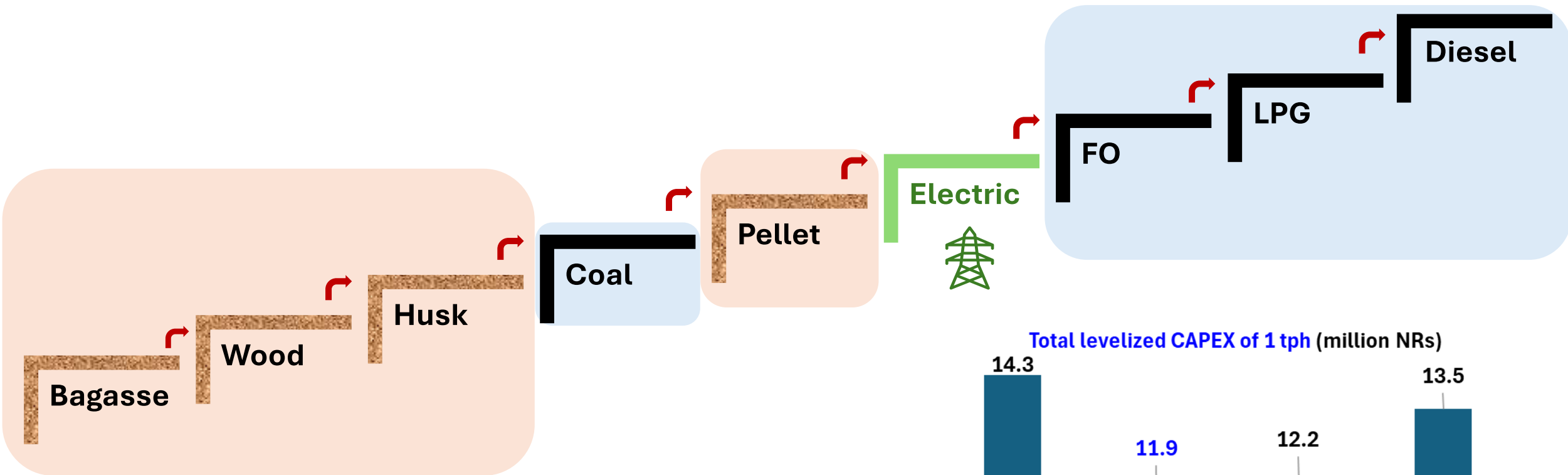
Total levelized cost of 1 tph boiler (million NRs)



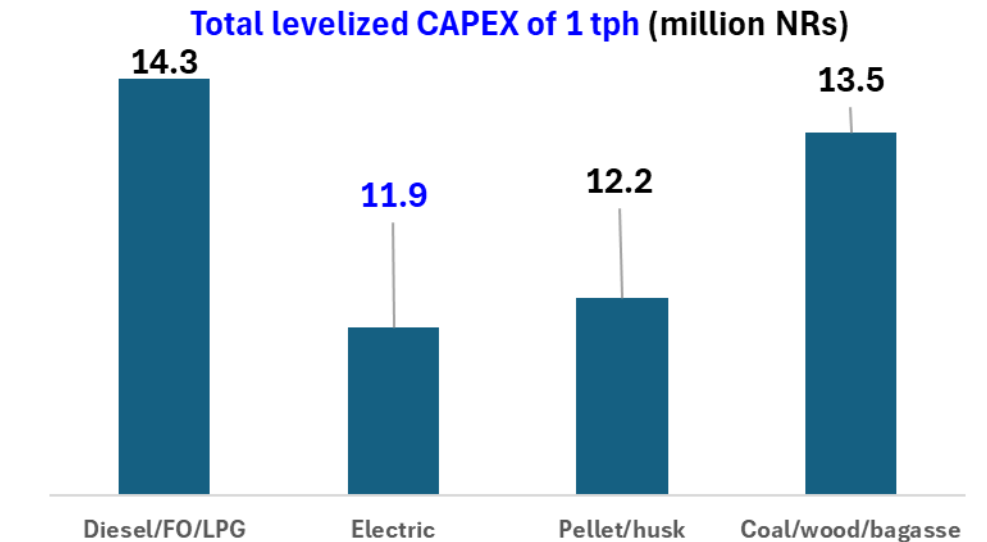
Note: * Others include installation, maintenance, insurance and contingency. CAPEX for all boilers except electricity includes pollution control equipment (PCE).

- **Electric** boilers are **cheaper** than **FO**, **LPG** and **diesel**, but more **expensive** than **coal**, and **biomass** boilers.
- **Fuel cost** is the **major** component of **boiler cost (39% to 86%)**.

Levelized CAPEX cost (Base case)



Levelized CAPEX cost of electric boiler is cheaper than biomass and fossil-fuel based boilers.



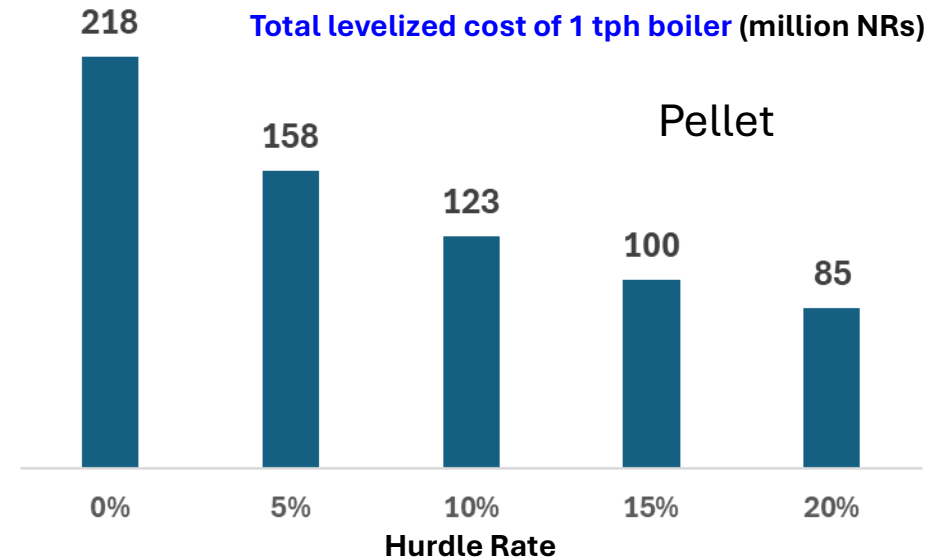
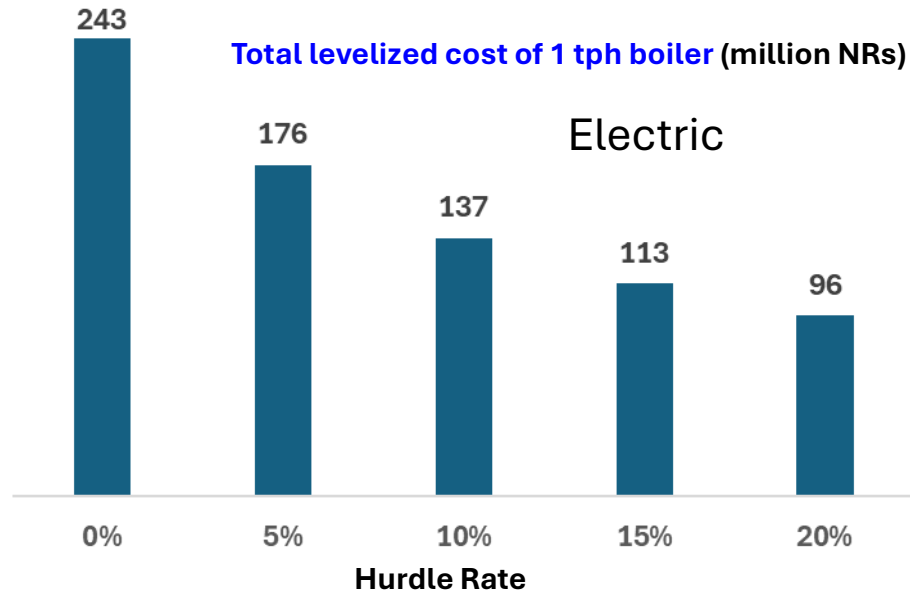
CAPEX includes both boiler equipment and pollution control equipment (PCE)

Levelized Total Cost (with Hurdle rate)

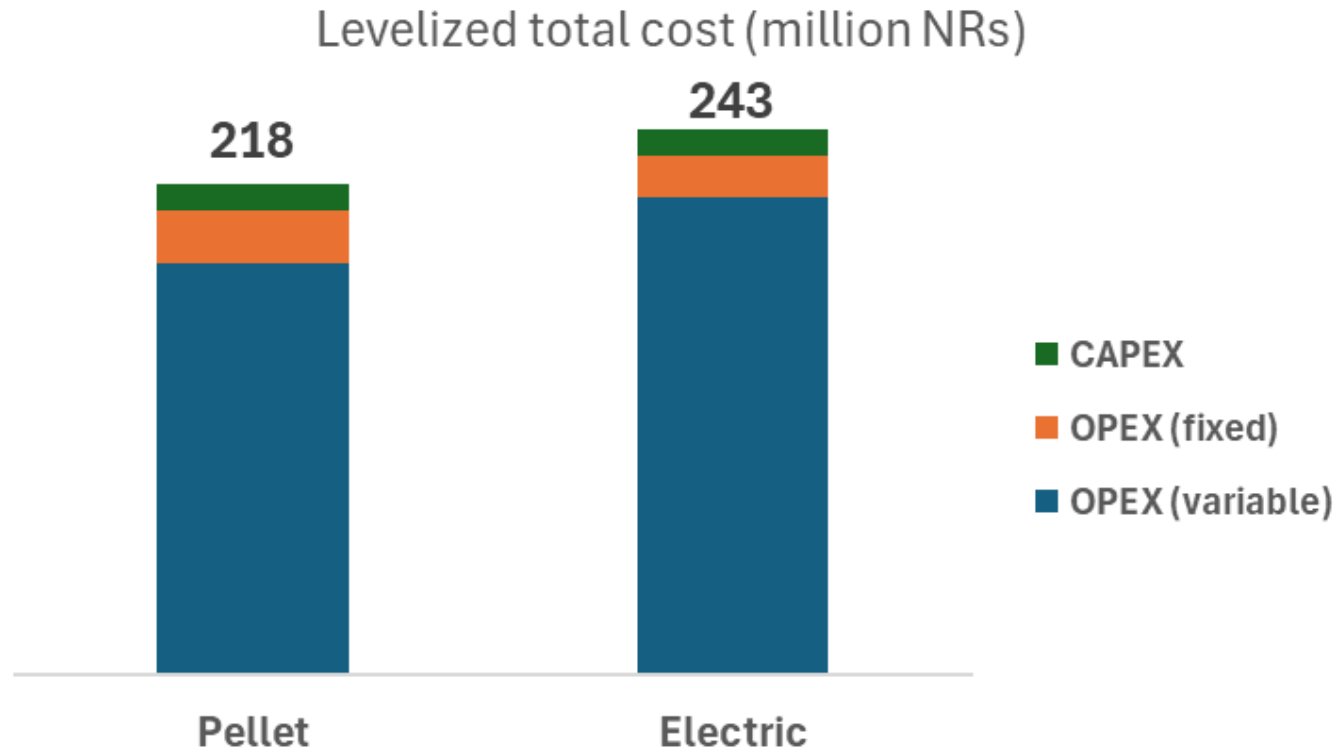
Total levelized cost of 1 tph boiler (million NRs)

Hurdle rate	Diesel	LPG	FO	Electric	Coal	Pellet	Husk	Wood	Bagasse
0%	416	303	289	243	216	218	143	126	94

Note: Andersen et al. (2020) reported implicit discount rates (a type of hurdle rate) for energy efficiency retrofits ranging from 10% to 32%. We apply a 20% hurdle rate to the electric boiler and a 10% hurdle rate to the pellet boiler, in addition to the standard 6% discount rate.



Total levelized cost (Pellet vs Electric)



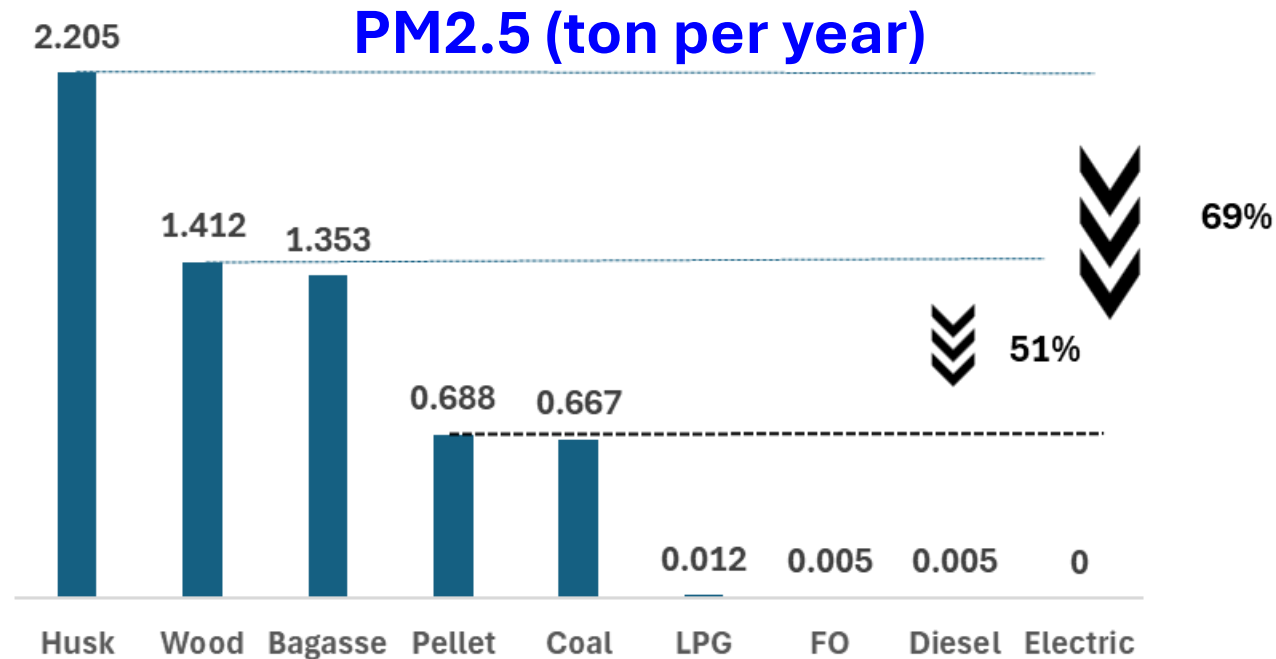
The **total levelized cost** of an **electric** boiler is **higher** than a **pellet** boiler.

Note:

OPEX (fixed) includes labor insurance and contingency. OPEX (variable) includes water, fuel, and maintenance.

Assumptions: Interest rate (8%), Loan term (10 years), Discount rate (6%), Inflation rate (5.5%). No grant and no down payment.

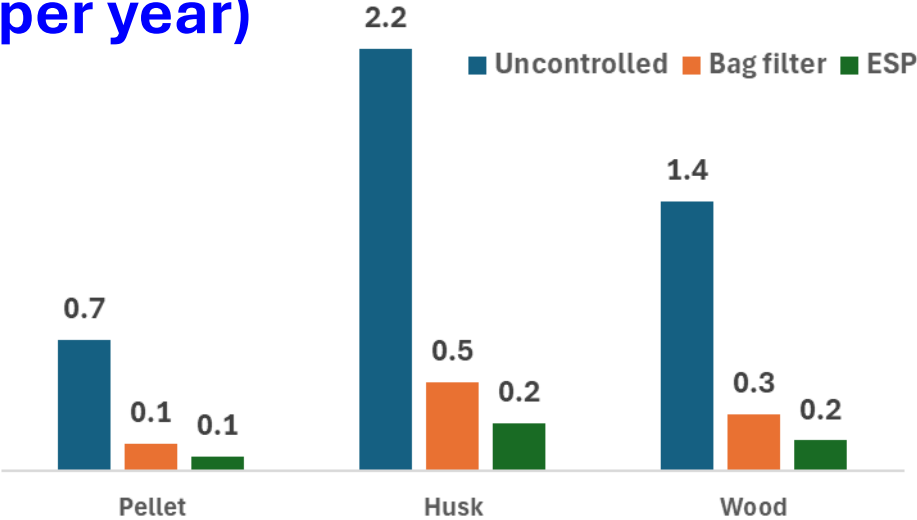
Estimated annual air pollutant emissions from the operation of 1tph boiler in Nepal (Base case)



- **PM2.5** emission is much **lower** with **pellets**.
- About **69% lower** than **husk** and **51% lower** than **wood**.
- **PM2.5** emissions are mostly from **biomass** and **coal**.
- **Husk** contributes most to **PM2.5**.

Average PM2.5 emissions of a pellet and other biomass boilers (1tph capacity) with and without PCE

PM2.5 (ton per year)



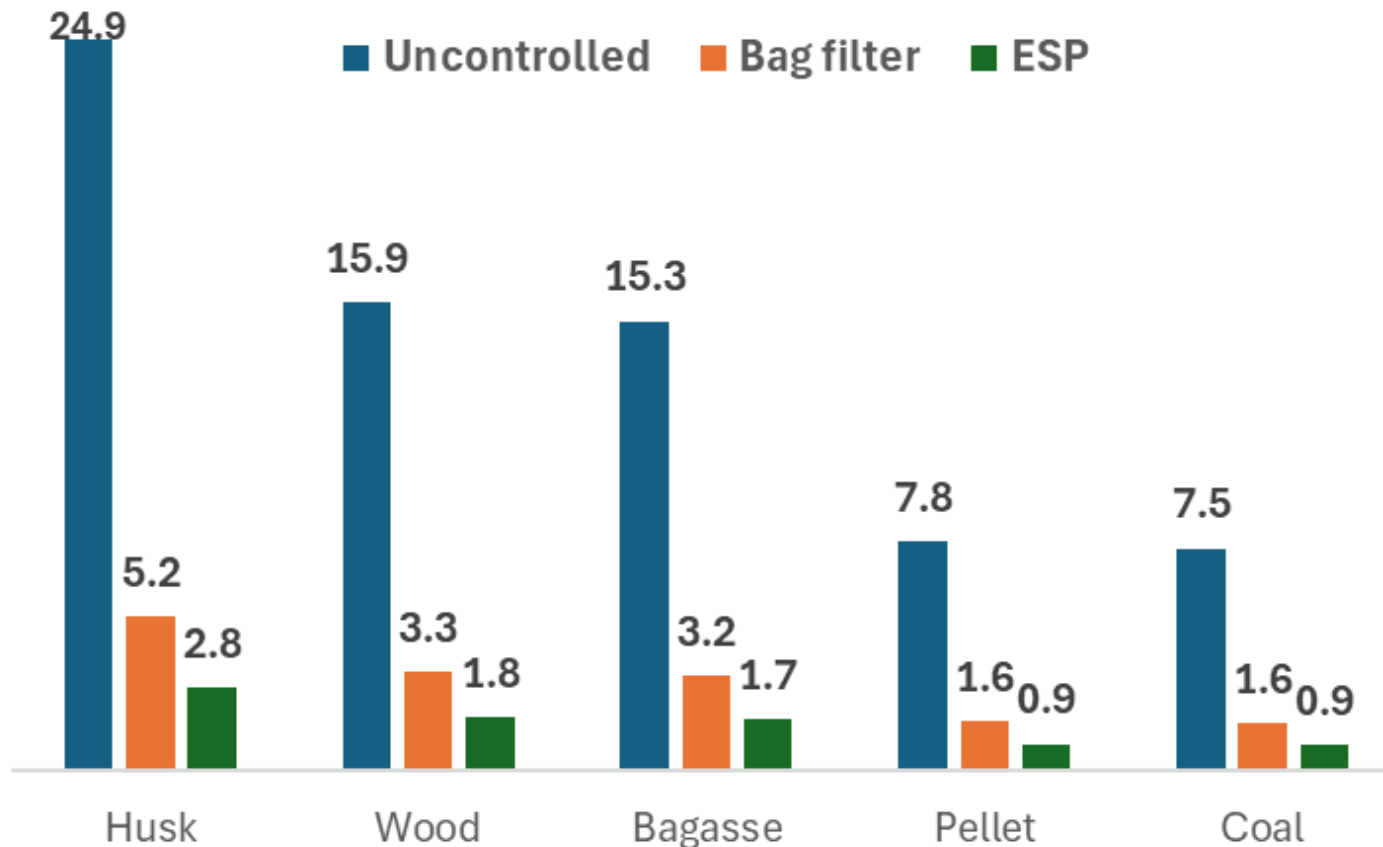
- With pollution control equipment (**PCE**), **PM2.5** can be reduced from **79%** (bag filters or wet scrubbers) to about **89%** (ESP) for **pellet** and **other biomass boilers** (e.g., husk and wood).

Biomass* boiler (1 tph capacity)	Without PCE (uncontrolled)	With bag filter	With ESP
PM 2.5 Reduction	0%	~ 79%	~ 89%

Note: * Based on dry wood (< 20% moisture content). PCE is pollution control equipment; ESP is an electrostatic precipitator.
Source: US EPA (2022). https://www.epa.gov/system/files/documents/2022-03/c1s6_fi-l_0.pdf for emission factors.

Health impact cost of PM2.5 emissions from boilers (1tph capacity) with and without PCE

Health impact cost of PM2.5 (million NRs per year)



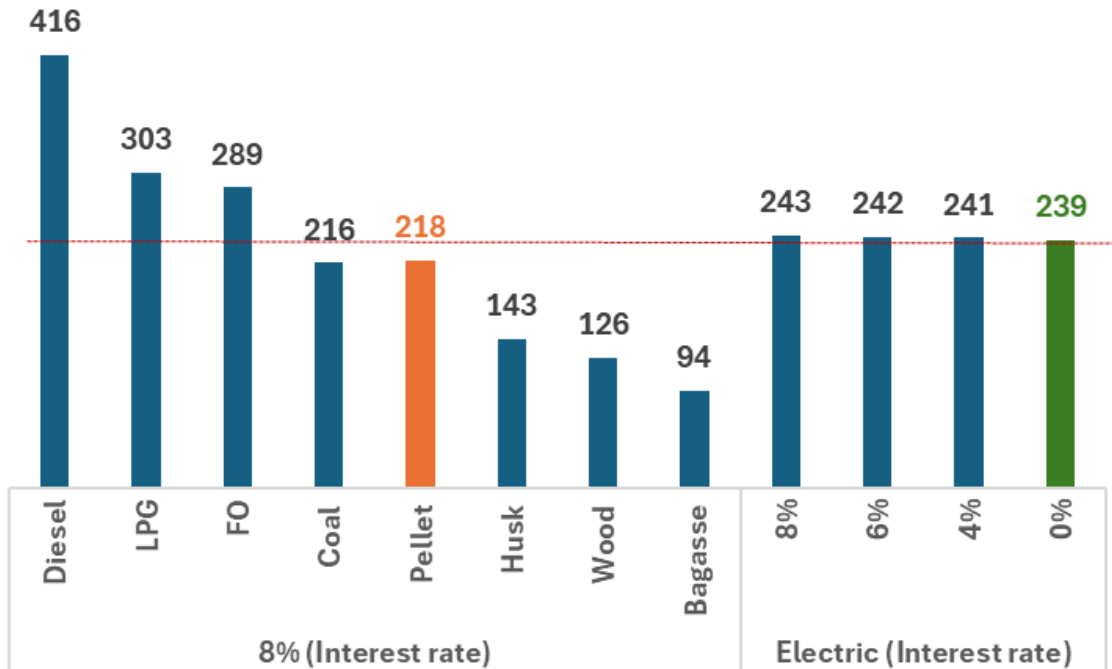
- Even with pollution control equipment (e.g., bag filters/ESP), there is a health impact cost of **PM2.5 emissions**, while an electric boiler has **zero** cost.

Sce-rio Cases

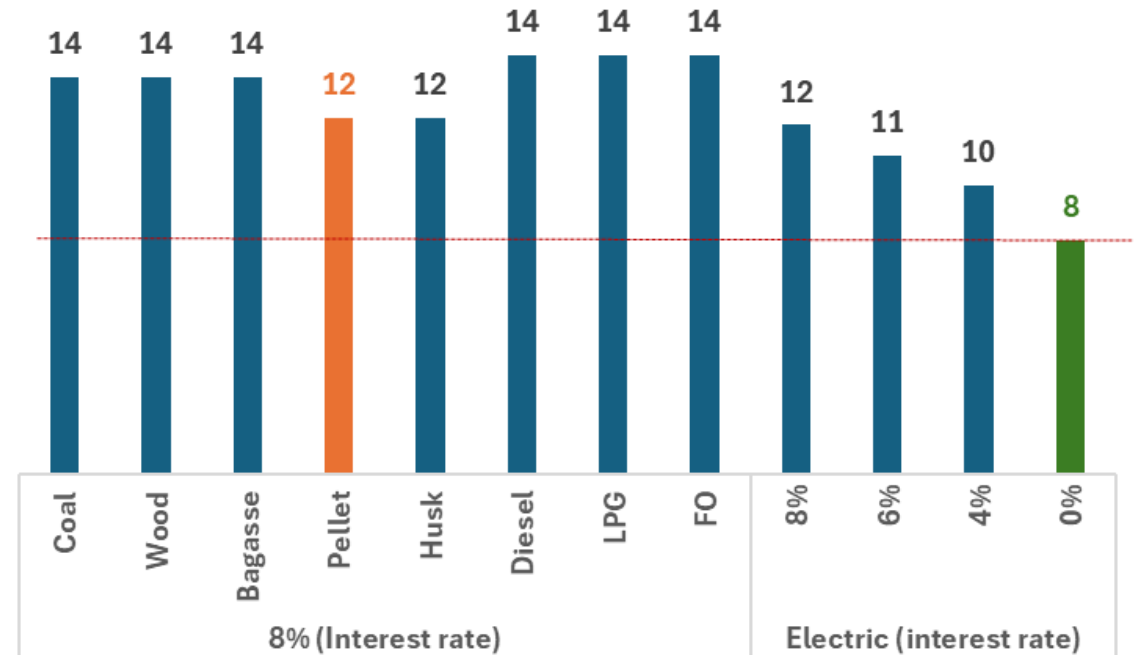
- **Capital subsidy (Electric)**
- **Production loan**
- **Increase in fossil fuel price**
- **Pellet promotion**

Impact of Capital Subsidy (Interest rate)

Total levelized costs (million NRs)

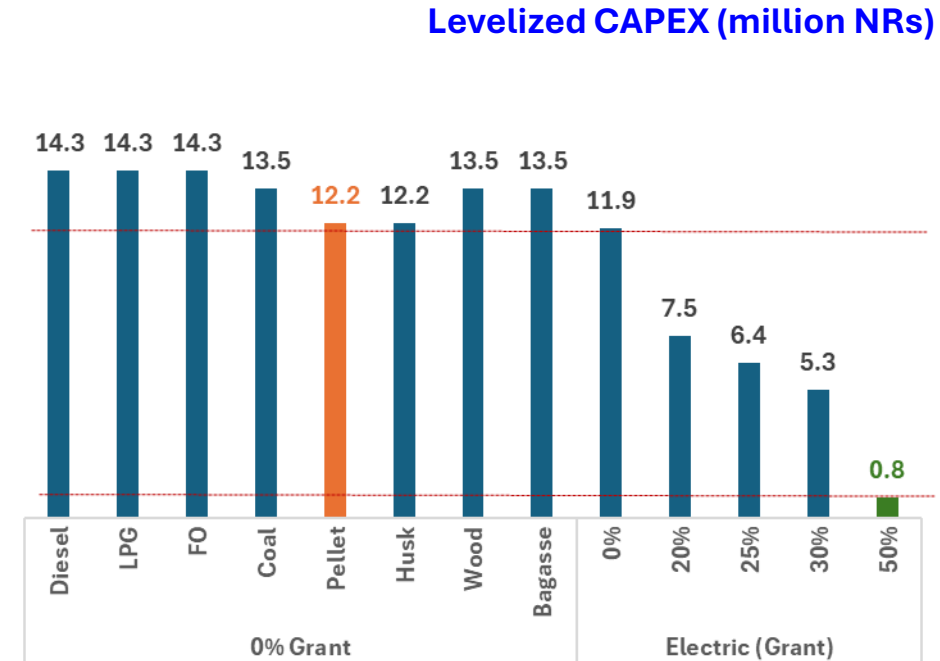
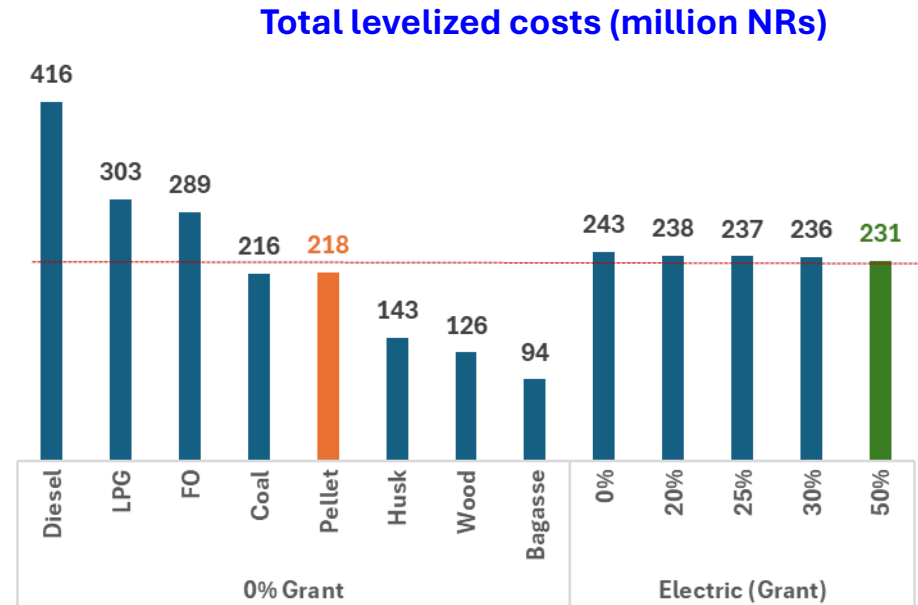


Levelized CAPEX (million NRs)



- Even with **0% interest** loans, the **electric** boiler still ranks as the **fourth most expensive** in total levelized costs.
- With a **0% interest** rate, levelized **CAPEX** is reduced by **one-third. Cheapest.**

Impact of Capital Subsidy (Grant)

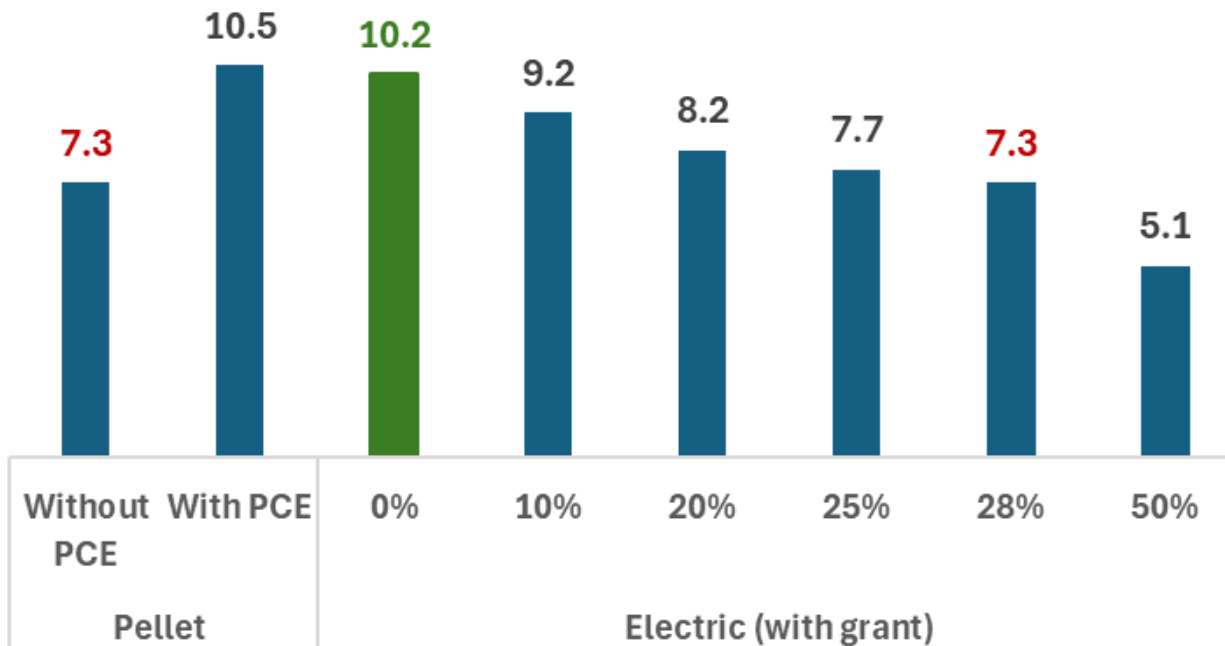


Even with a **50%** capital grant, **electric** boilers are still more **expensive** than **pellet, coal, and other biomass fuels**, but **cheaper** than **oil-based** fossil fuel boilers in terms of total levelized cost.

When **only levelized CAPEX** is considered, even with **0%** grant, the **electric** boiler becomes the **cheapest**.

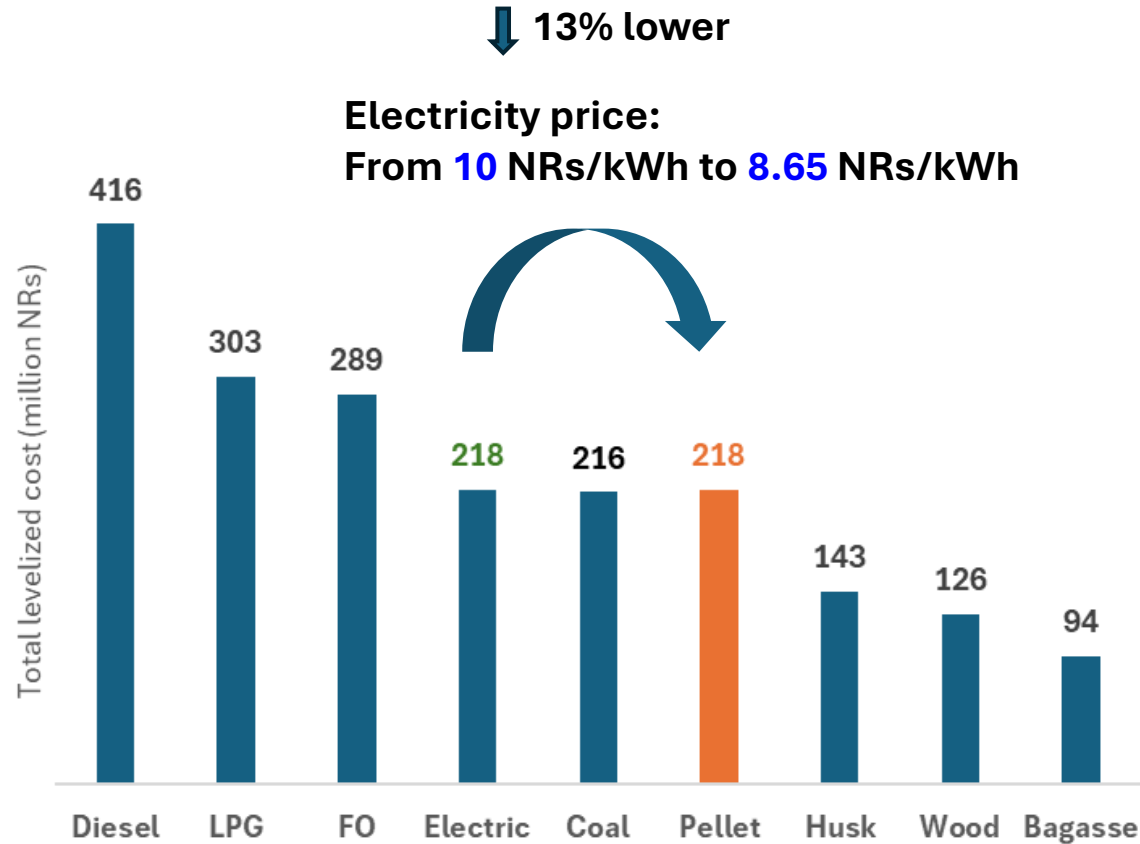
Capital cost with a grant for an electric boiler compared with a pellet boiler

Capital cost of boiler with grant for electric (million NRs)



- Even with a **0 % grant**, an **electric** boiler is **cheaper than** a **pellet** boiler **with PCE**.
- Without PCE, a **28% grant** is required for the **electric** boiler to make it **competitive** with the **pellet** boiler.

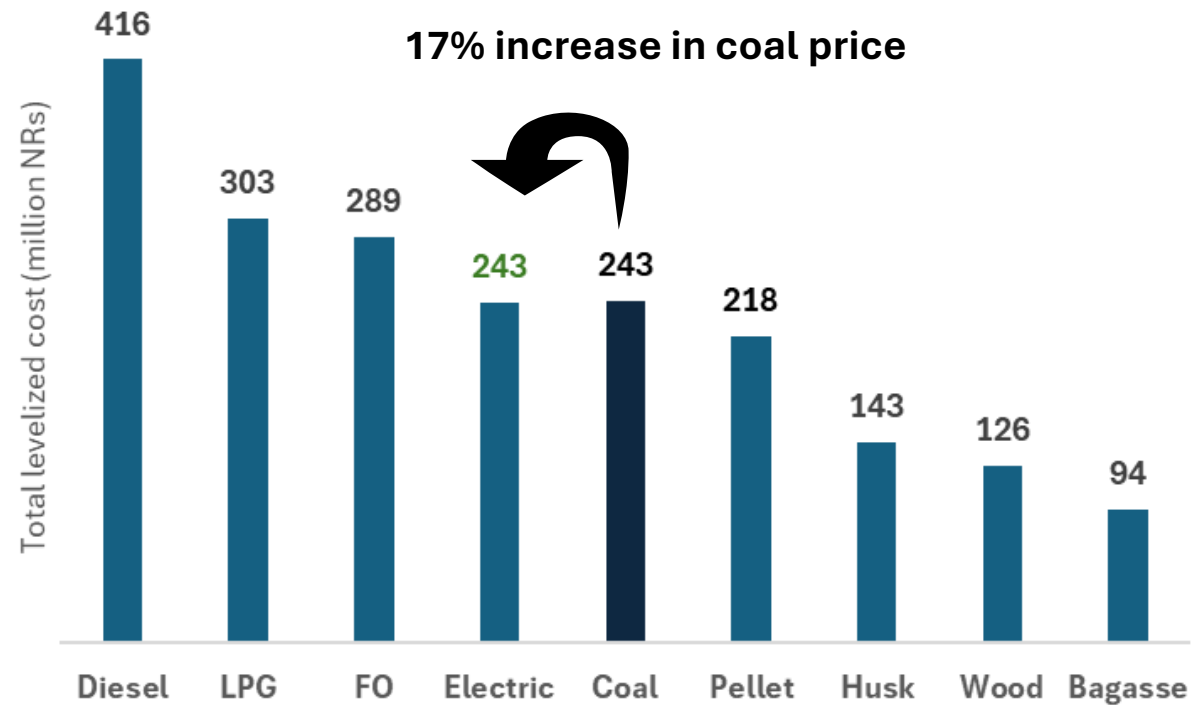
Production loan (electricity price) to make the electric boiler attractive with a pellet boiler



13% electric price reduction is needed from the base case (10 NRs/kWh) to equalize the electric boiler with the **pellet** boiler.

Average electricity **selling price** to **India** (2024): 7.6 NRs/kWh

Production loan (coal price) to make the electric boiler attractive with a coal boiler

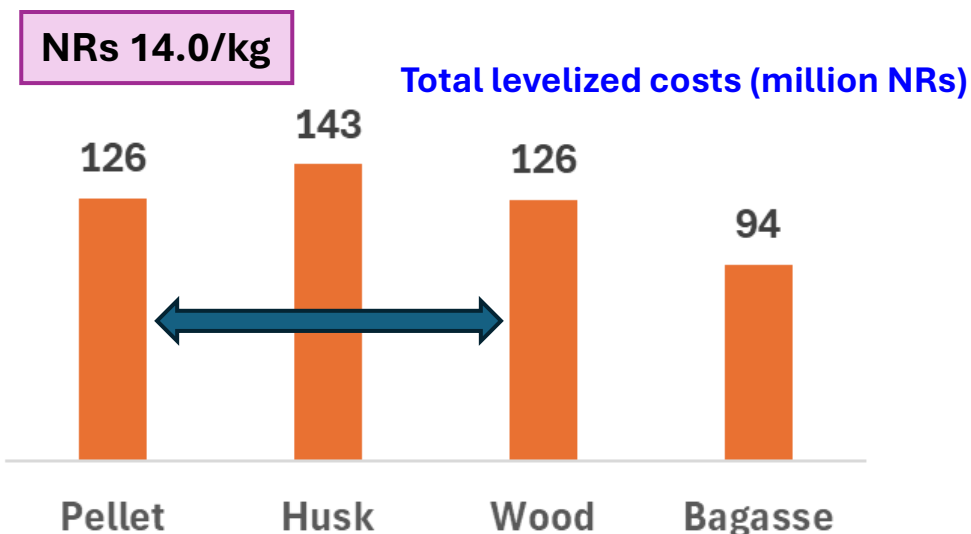
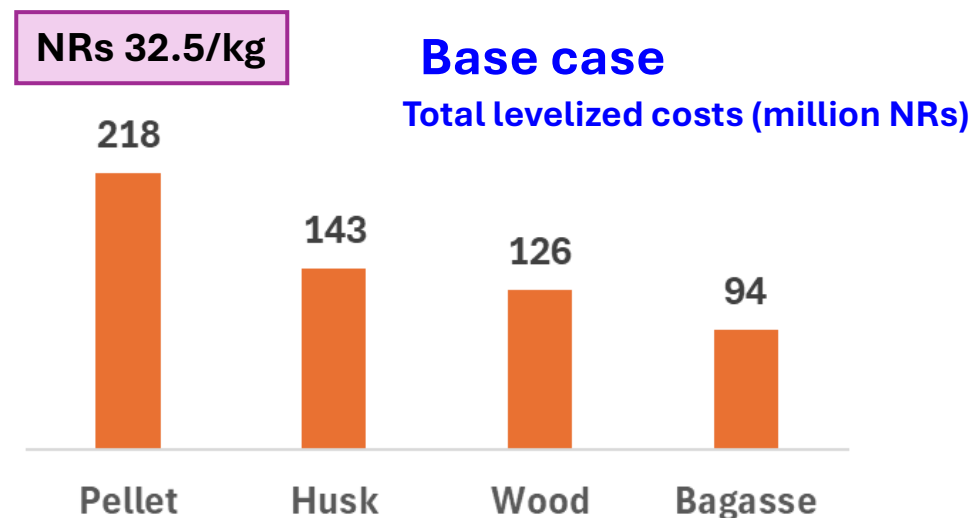
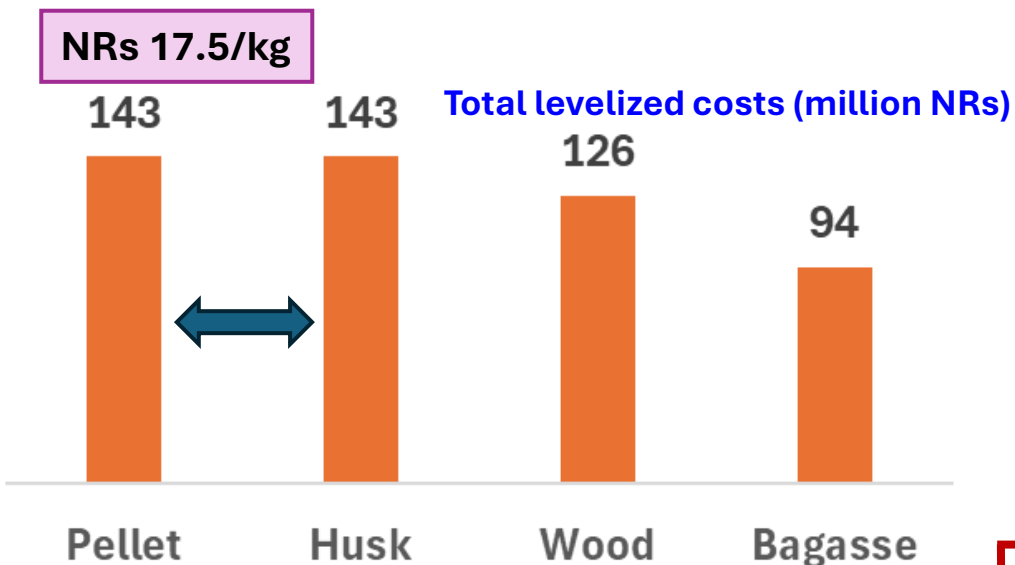


Fuel price (NRs/kg)

	Base case	New	% Ch
Coal	40	46.8	17%

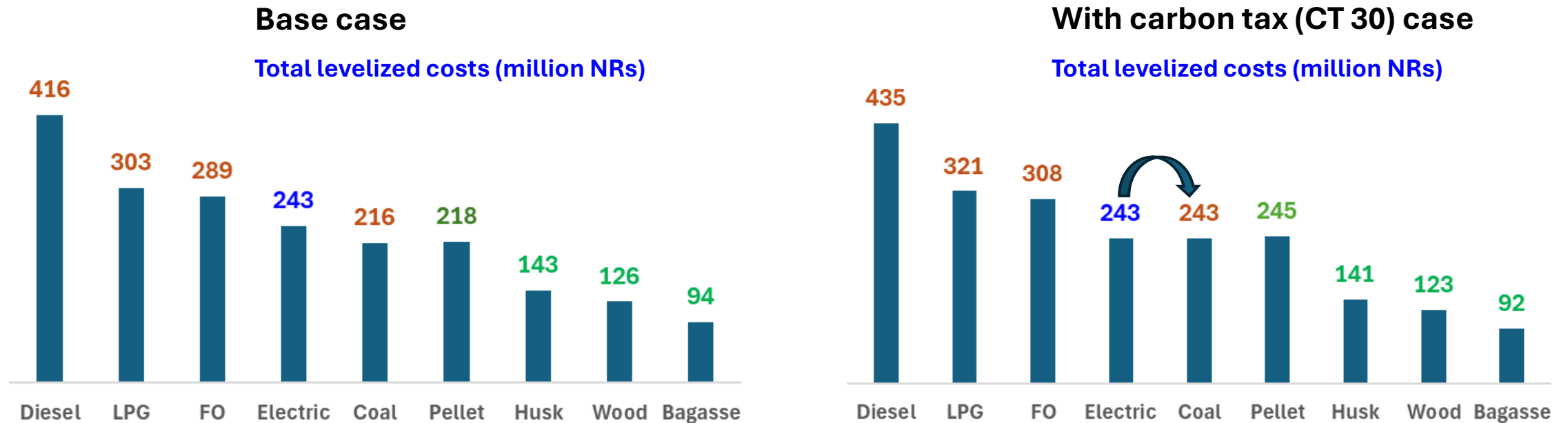
Increase in fuel price of **coal** to match with **electric**.

Pellet promotion (Reduction in pellet price)



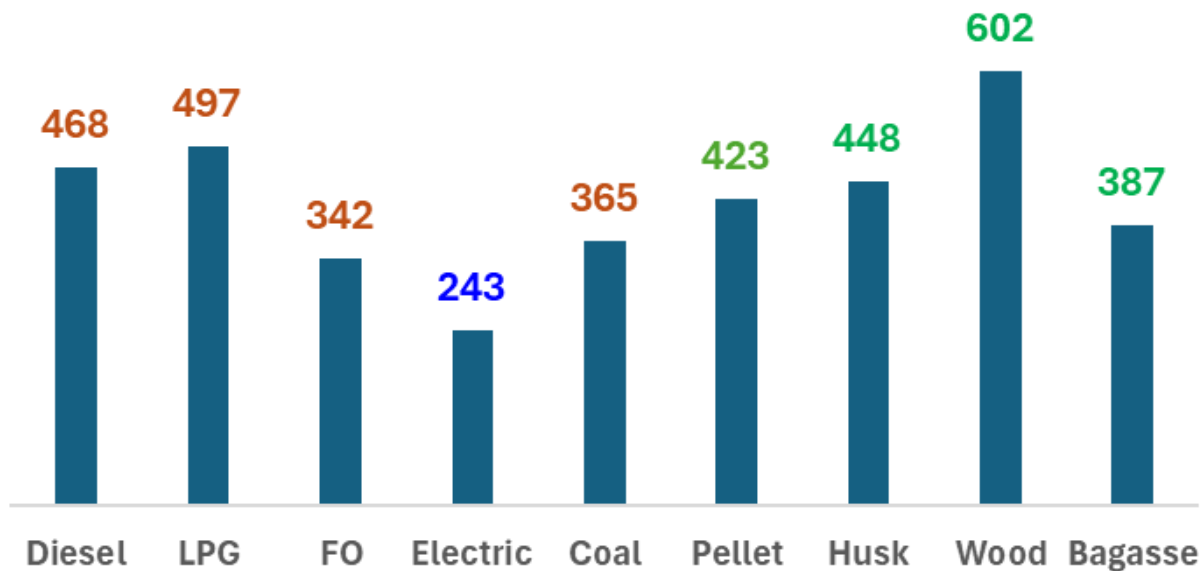
- **46% price reduction** from the current price to make **pellet** boiler competitive with **husk**.
- **57% price reduction** to make **pellet** boiler competitive with **wood**.
- Pellets are primarily **imported** from **India**.
- This also **incentivizes** the promotion of **domestic pellet production**.

Production loan (green tax) to make the electric boiler attractive with a coal boiler



If the coal price increases by **17%** (roughly about carbon tax of USD **30** per tCO₂), the **electric** boiler becomes competitive with the **coal** boiler.

Total annualized cost with environmental damage cost (million NRs)



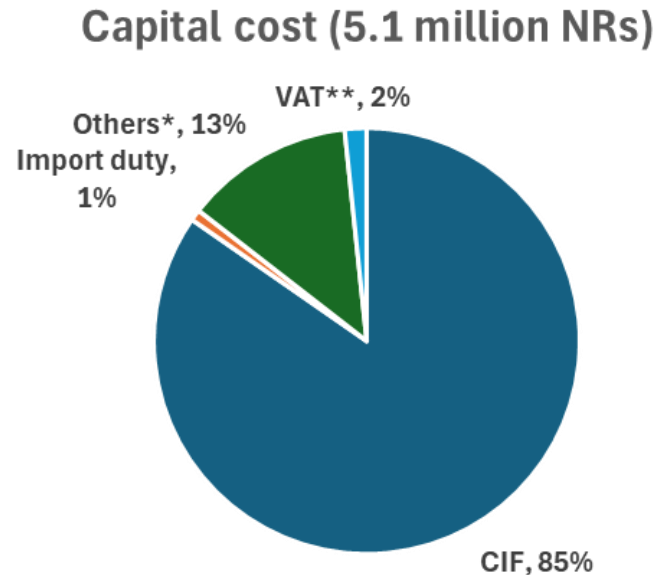
Note: The social costs of CO₂ are USD 0.04/kg of CO₂, and PM_{2.5} is USD 86.85/kg of PM_{2.5}.
Source: EPIC India (2018).

- Environmental damage costs
 - **Environmental** damage cost (CO₂)
 - **Health** damage cost (PM_{2.5})
- With these costs, **Electric** boilers are the most **economical** one.
- **Biomass** boilers are relatively more **expensive** compared to **coal** and **FO** boilers.

Electric pelletization (Machine) Promotion

Electric Pellet Production Machine

Electric Pellet machine



* Includes transportation, S&H, and other civil works.

** Only applicable for others.

Estimated production cost and selling price of domestic pellet by feedstock type (NRs/kg)

	Husk pellet (NRs/kg)	Wood pellet (NRs/kg)
Production cost*	14.29	20.80
VAT (13%)	1.86	2.70
Profit margin (15%)	2.15	3.12
Sell price	18.30	26.62

*Based on average cost and annual production of 2,520 tons of pellets over the period of 20 years. The feedstock prices assumed are NRs 6/kg for rice husk (based on MoHA) and NRs 10/kg for wood. Production costs exclude land costs, transportation, and collection costs of feedstock.

Domestic pellet production (NRs 18.3/kg) is cheaper than **importing** it from India (NRs 32.5/kg).

Electric Pellet Production (Levelized ROI)

Return on investment (ROI) over 20 years with pellet selling price at NRs17.5/kg

	Unit	Feedstock		
		Husk	Sawdust	Bagasse
Total discounted operating cost	(10^6 NRs)	403	362	480
Total discounted output	(ton)	29,047	29,047	29,047
Levelized cost of production	(NRs/kg)	14	12	17
Selling price (revenue) per kg	(NRs)	17.5	17.5	17.5
Profit per kg	(NRs)	4	5	1
Total levelized profit	(10^6 NRs)	106	219	28
Total CAPEX (pv)	(10^6 NRs)	6	6	6
ROI over 20 years	(%)	1,797	2,484	480
Average ROI per year	(%)	90	124	24

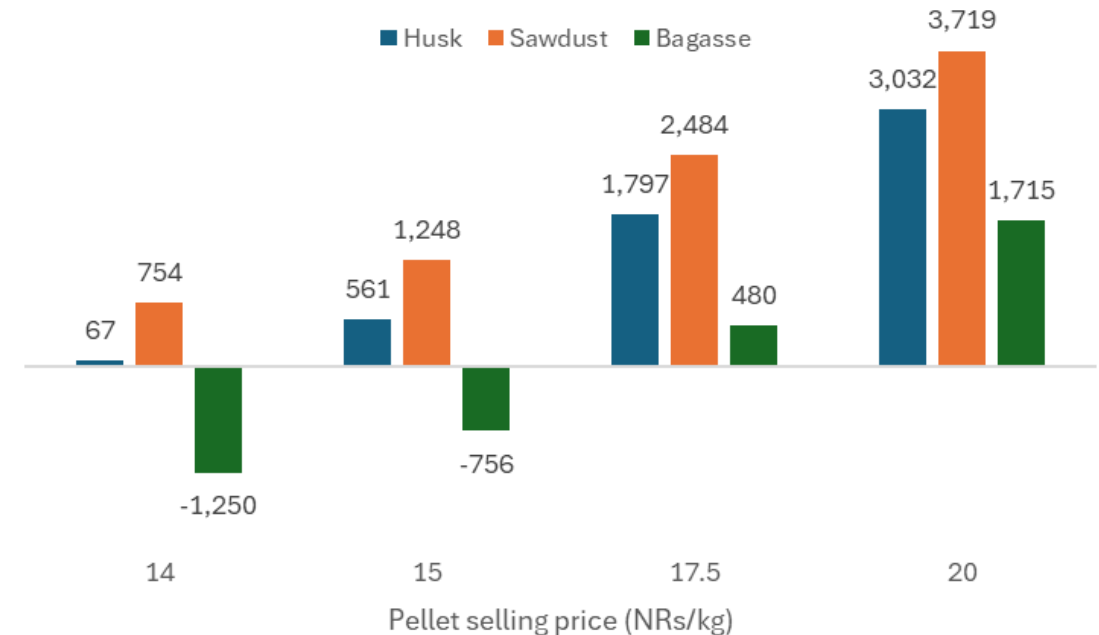
Assumptions:

Lifetime 20 years, discount rate: 6%, and pellet selling price is NRs 17.5/kg.

Production costs exclude land costs, transportation, and collection costs of feedstock.

$$ROI_{Levelized}(\%) = \frac{Total\ lifetime\ levelized\ profit_{pv}}{Total\ CAPEX_{pv}} \times 100$$

Return on investment (ROI) over 20 years with different pellet selling prices (sensitivity)



Domestic pellet production with **NRs 17.5/kg or higher** as a selling price, the investment on electric pellet machine is financially profitable for all feedstocks (husk, sawdust, and bagasse).

Thank You.

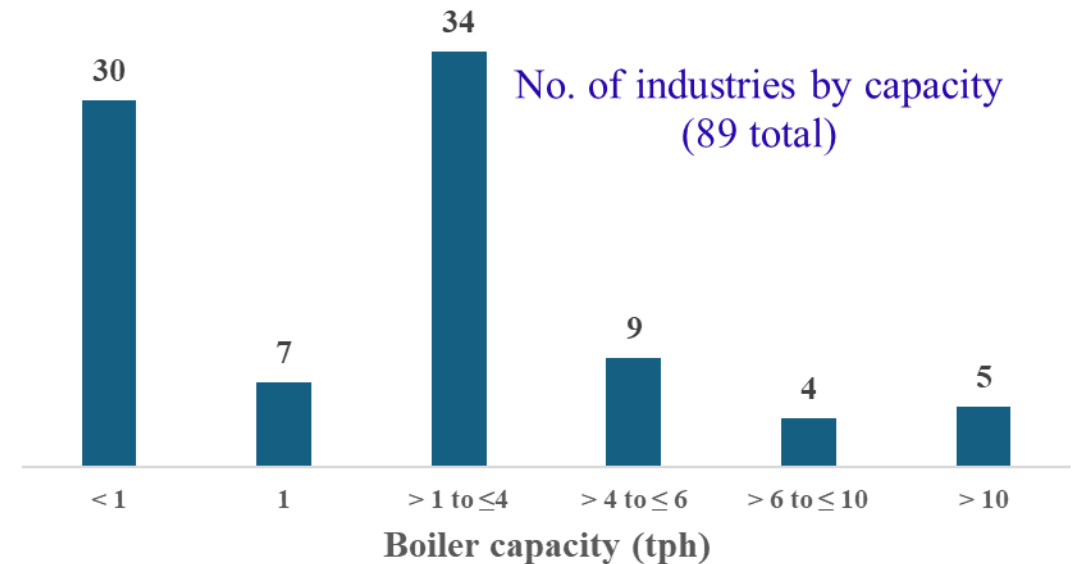
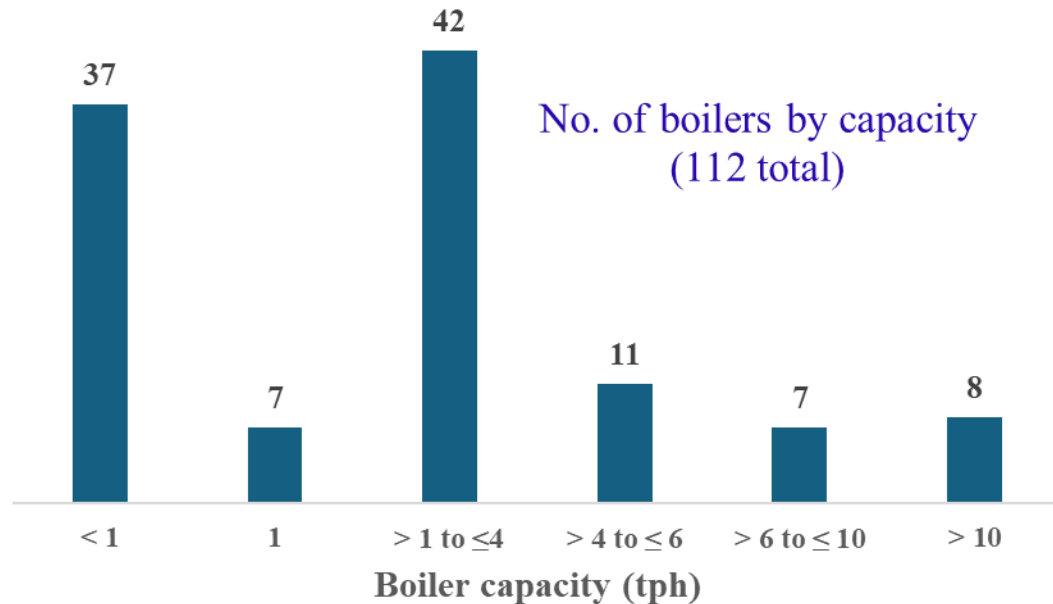
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Extra survey summary slides

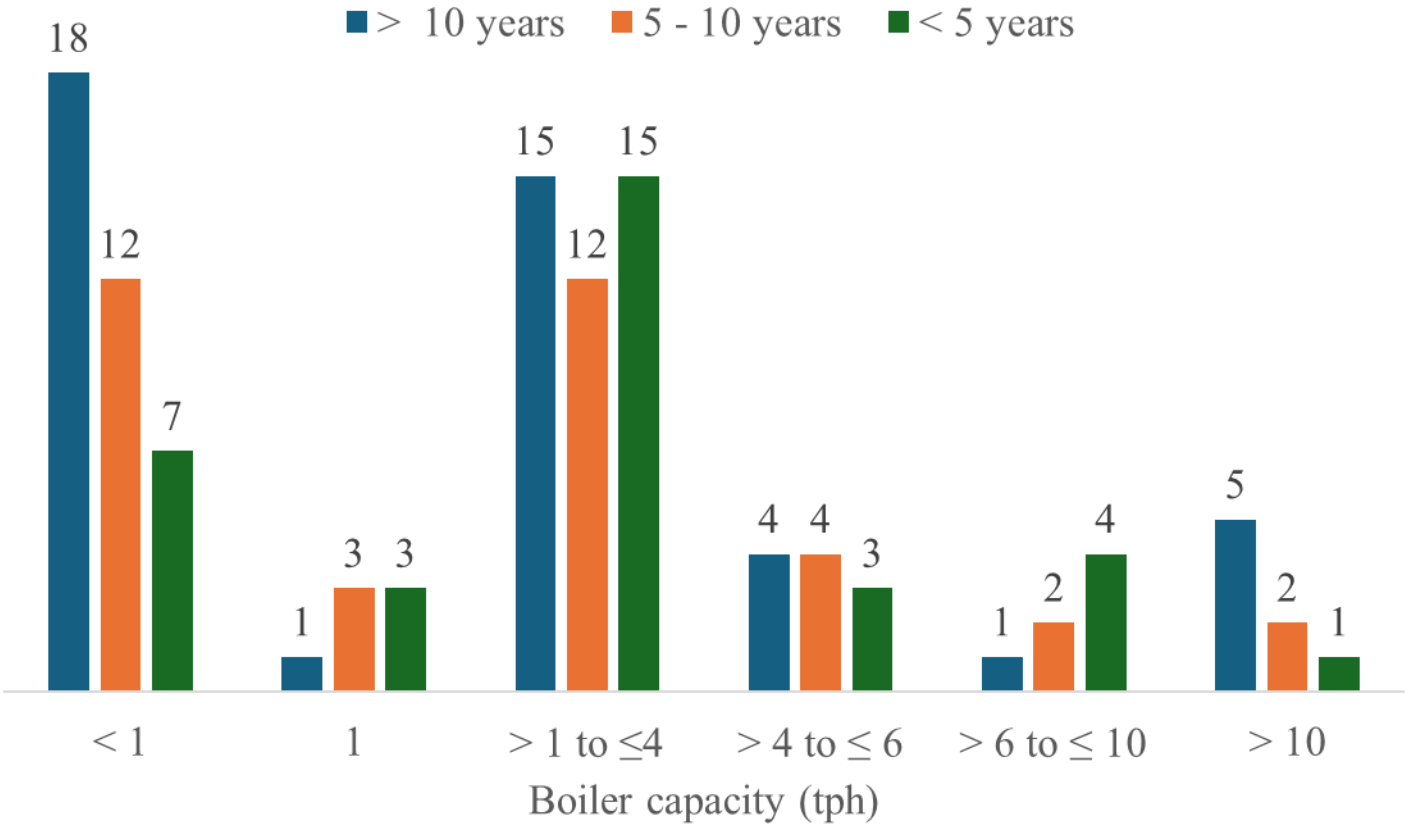


Firm Survey

Distribution of boilers and industries by boiler capacity



Distribution of boilers by boiler capacity



Boiler age	Total
> 10 years	44
5 - 10 years	35
< 5 years	33

No. of boilers by type of fuel and capacity

Capacity (tph)	Bagasse	Briquette	Coal	Diesel	Electricity	Firewood	FO	LPG	Pellet	Husk	Total
< 1				15		18		1	2	1	37
1		1		1	1	4					7
> 1 to ≤ 4				6		15	2		1	18	42
> 4 to ≤ 6	1					3				7	11
> 6 to ≤ 10						3				4	7
> 10			1							7	8
Total	1	1	1	22	1	43	2	1	3	37	112
% sh of total	<1	<1	<1	20	<1	38	2	<1	3	33	